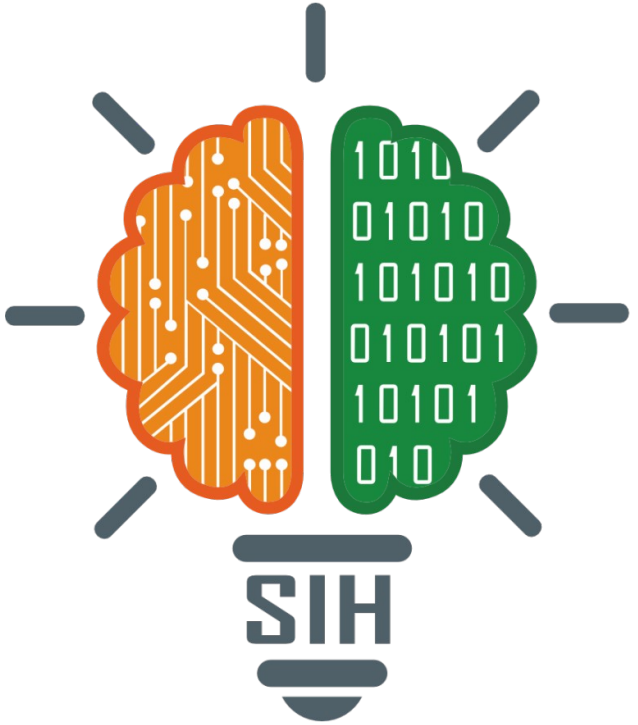
SMART INDIA HACKATHON 2026

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PRAHARI — The Countdown Band

Warns before the emergency — the flood, cyclone or heatwave coming at you, and the heat strain building inside you — and reaches workers no tower reaches.



- **Problem Statement ID – 26181**
- Problem Statement Title – A secure, AI-powered Personal Health Companion that delivers real-time, privacy-preserving health monitoring and early warning capabilities, helping individuals recognize health risks before they become emergencies. The solution should improve resilience during heat waves, floods, pollution events, and other disasters common in India while enabling continuous health support through on-device intelligence.
- **Theme – MedTech / BioTech / HealthTech**
- **PS Category – Hardware**
- **Team ID – <your team id>**
- **Team Name – Point Break**

PRAHARI: two warnings, one system.

Proposed solution / Personal heat-strain awareness + official multi-hazard early warning



THE HAZARD OUTSIDE

Flood, cyclone, heatwave and lightning alerts can fail to reach the last mile.



THE STRAIN INSIDE

Heat exposure and incomplete recovery differ from one worker to the next.

01 THE BAND

- Heart rate, skin temperature, ambient heat and motion
- Local risk trend, OLED status and distinct haptic patterns
- LoRa peer relaying extends alerts beyond the gateway

02 THE COMPANION APP

- Fetches official NDMA SACHET CAP alerts when online
- Sends relevant alerts by BLE to a LoRa gateway band
- Keeps offline history and recovery records on the phone

INNOVATION & UNIQUENESS

Personal countdown*

Estimate time to a heat-risk threshold only when signal quality permits.

Thermal debt ledger*

Carry cumulative strain and recovery across shifts and overnight.

Two warnings, one wrist

Separate hazard and strain patterns; local reception needs no personal phone.

*Proposed, to validate. Photos: Dibakar Roy (Kolkata flood); Abhishek Gupta (Indian farmer) / Pexels. Context images, not trials.

System architecture / two inputs, one wrist.

Proposed end-to-end flow · body processing stays local; official alerts enter through a connected gateway

A / PERSONAL HEAT-STRAIN PATH

SENSOR INPUTS
MAX30102 · heart rate
MAX30208 · skin temp
AHT20 · air temp / RH
IMU · motion

→

XIAO nRF52840
Filter + sample signals
Calculate signal quality
C/C++ on the band

→

Usable signal?

YES →

ESTIMATE + FORECAST
Kalman + personal baseline
Trend to threshold*
Recovery ledger*

→

LOCAL WARNING
OLED risk / time*
Strain vibration
Prompt to take action

NO ↓

Estimate unavailable · refit / resample

B / OFFICIAL HAZARD-ALERT PATH

NDMA SACHET
Official CAP / RSS alerts

→

PHONE APP
Fetch · location · expiry

→

GATEWAY BAND
BLE in → SX1262 LoRa

→

PEER BANDS
Deduplicate · hop limit

→

HAZARD WARNING
Distinct wrist vibration

PLANNED STACK C/C++ · Kalman / TinyML evaluation · Flutter + local storage · protected keys · raw physiology kept local

*Validation required. Low-quality data must not imply safety. New CAP alerts need connected ingress. Air temp + RH alone are not full WBGT.

FEASIBILITY AND VIABILITY

From a working sensing core to a supervised pilot.



COMPACT, MODULAR HARDWARE

XIAO + sensor modules
Two-layer rev B carrier PCB
LiPo protection + haptic output

VERIFIED

Live PPG / heart rate, OLED and I²C diagnostics.

IN PROGRESS Radio bring-up; PCB checks; app + forecast integration.

CHALLENGES → MITIGATION

Motion / uncertainty	Quality gating + reference-device testing; no false reassurance.
Radio outages / old alerts	Expiry, deduplication and hop limits; measure delivery and range.
Battery / fit / robustness	Duty-cycle sensing and radio; test comfort, sweat and shift use.
Data protection	Keep physiology local; validate access control and key handling.

ECONOMIC & OPERATIONAL VIABILITY

Off-the-shelf parts simplify repair and sourcing. Quote the full rev B assembly, enclosure and battery before setting a device price.

IMPLEMENTATION GATES

01 Close hardware and radio issues

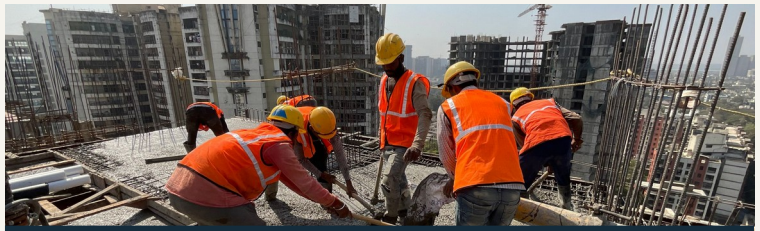
02 Validate sensing, relay and power

03 Supervised pilot and user feedback

Photo: Seeed Studio, actual XIAO module. Bench results do not establish clinical accuracy or field readiness; full-device cost remains to be quoted.

IMPACT AND BENEFITS

Put the warning where action can start.



OUTDOOR WORK CREWS



FARMING COMMUNITIES



DISASTER-EXPOSED AREAS

57% **of Indian districts at high to very high heat risk**
CEEW, 2025 [1]

SOCIAL

Distinct haptics support simple warnings.
No personal phone needed for reception.
Local crews can act on shared alerts.

ECONOMIC

Timely rest prompts aim to reduce disruption.
Crew deployment can share gateway access.
Measure time to action before claiming savings.

SYSTEMIC & ENVIRONMENTAL

Extend official alerts into local peer networks.
Rechargeable, modular design aims for repair.
Validate durability and lifetime in the pilot.

PILOT METRICS Delivery time • Missed / false alerts • Time to action • Battery per shift • Comfort and comprehension

Real photographs: yajna S R; Abhishek Gupta; Dibakar Roy / Pexels. Intended benefits, not measured health or economic outcomes.

Research foundations. Evidence. Next questions.

01 **CEEW · Extreme heat in Indian districts (2025)**

Heat-risk evidence and target-user context.

02 **NDMA · SACHET CAP / RSS alerts**

Official alert source for hazard dissemination.

03 **Buller et al. · Wearable physiological estimation**

Kalman-filter research; not PRAHARI validation.

04 **Seeed Studio · XIAO nRF52840 documentation**

MCU, BLE, Sense IMU and component image.

05 **PRAHARI · firmware, KiCad and project notes**

Sensor/OLED work, BOM and bring-up evidence; private repo.

WHAT THE PILOT MUST TEST

01 Does the forecast stay useful during movement and recovery?

02 Do alerts arrive reliably across the actual deployment terrain?

03 Can users distinguish, trust and act on the warning?

Obsidian brain + Idea_3 PDF
+ project report informed this deck.

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yajna S R — Mumbai workers | Abhishek Gupta — Indian farmer | Dibakar Roy — Kolkata flood | Seeed Studio — XIAO